

Introduction to Python

What is Python?

Python is a **high-level, interpreted, and general-purpose programming language**. It is easy to learn because of its simple syntax, which is similar to the English language. Python is widely used in:

- Web Development
- Data Science
- Machine Learning
- Artificial Intelligence
- Automation
- Game Development
- Software Development

Python is known for:

- Simple and readable code
 - Easy debugging
 - Large community support
 - Huge collection of libraries
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History of Python

Creator of Python

Python was created by Guido van Rossum.

He started working on Python in the late 1980s at Centrum Wiskunde & Informatica (CWI) in the Netherlands.

Why the Name “Python”?

The name **Python** was inspired by the British comedy show:

Monty Python's Flying Circus

It was not named after the snake.

Evolution of Python

Version	Year	Features
Python 1.0	1994	Basic features introduced
Python 2.0	2000	Added garbage collection and Unicode support
Python 3.0	2008	Improved syntax and removed old features
Python 3.x	Present	Faster, secure, and widely used

Features of Python

1. Easy to Learn

Python syntax is simple and understandable.

Example:

```
print("Hello World")
```

2. Interpreted Language

Python code is executed line by line using an interpreter.

3. Platform Independent

Python programs can run on:

- Windows
- Linux
- macOS

without changing the code.

4. Object-Oriented Language

Python supports concepts like:

- Class
 - Object
 - Inheritance
 - Polymorphism
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5. Open Source

Python is free to use and download.

Official website: [Python.org](https://python.org)

6. Large Library Support

Python provides many built-in libraries such as:

- NumPy
 - Pandas
 - Matplotlib
 - TensorFlow
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Applications of Python

Web Development

Frameworks:

- Django
- Flask

Data Science

Libraries:

- Pandas
- NumPy

Machine Learning & AI

Libraries:

- Scikit-learn
- TensorFlow

Automation

Used for:

- File handling
- Web scraping
- Task automation

Game Development

Libraries:

- Pygame
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Advantages of Python

- Simple syntax
 - Faster development
 - Huge community support
 - Cross-platform support
 - Easy integration with other languages
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Disadvantages of Python

- Slower than C and Java
 - High memory usage
 - Not preferred for mobile app development
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Python Example Program

```
name = input("Enter your name: ")  
print("Welcome", name)
```

Output:

Enter your name: Karthik

Welcome Karthik

Compiler vs Interpreter

Compiler	Interpreter
Converts whole program at once	Converts line by line
Faster execution	Slower execution
Errors shown after compilation	Errors shown immediately
Example: C, C++	Example: Python

Python uses an **Interpreter**.